

KUUSAMO KIUTAKONGAS, Finland

WMO: 028110

Lat: 66.3700N

Lon: 29.3119E

Elev: 165

StdP: 99.36

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-34.2	-31.2	-38.0	0.1	-34.2	-34.7	0.1	-31.2	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.2	25.6	16.4	23.5	15.4	21.4	14.8	18.3	22.7	17.1	21.5	16.0	19.6	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.7	12.1	19.8	15.5	11.2	18.6	14.3	10.3	17.6	52.3	22.8	48.7	21.6	45.3	19.8	23.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) N/A	N/A	N/A	DB	-36.4	27.6	3.0	2.3	-38.5	29.2	-40.3	30.6	-41.9	31.9	-44.1	33.5	(4)
(5)			WB	-36.5	19.4	2.9	1.8	-38.6	20.7	-40.4	21.8	-42.0	22.8	-44.1	24.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	0.6	-14.0	-11.7	-7.6	-0.6	6.3	10.9	14.9	12.4	7.7	0.8	-4.3	-8.9	(6)	
	DBStd	11.06	8.29	8.83	6.33	4.60	4.54	4.08	3.62	3.38	3.32	4.12	5.77	7.84	(7)	
	HDD10.0	3787	743	608	545	317	134	38	4	14	82	284	431	585	(8)	
	HDD18.3	6501	1002	842	803	567	374	226	117	184	319	543	681	843	(9)	
	CDD10.0	342	0	0	0	0	18	64	156	90	14	0	0	0	(10)	
	CDD18.3	13	0	0	0	0	0	2	9	2	0	0	0	0	(11)	
	CDH23.3	196	0	0	0	0	13	31	121	32	0	0	0	0	(12)	
	CDH26.7	28	0	0	0	0	1	3	18	6	0	0	0	0	(13)	
(14) Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(14)	
(15) Precipitation	PrecAvg	554	36	30	28	29	46	63	80	62	51	48	42	39	(15)	
	PrecMax	722	70	80	50	43	77	127	143	139	97	92	93	69	(16)	
	PrecMin	439	16	6	8	10	20	15	28	24	10	8	9	8	(17)	
	PrecStd	78	15	17	12	11	16	27	26	26	25	22	19	15	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.5	3.4	6.4	15.3	25.1	26.9	28.7	27.1	19.5	10.8	6.0	2.3	(19)	
		MCWB	1.0	2.0	2.6	7.9	13.4	15.9	17.6	19.3	14.1	9.7	5.3	1.5	(20)	
	2%	DB	-0.2	1.6	3.5	10.0	21.8	23.4	26.5	23.3	16.9	8.7	4.3	1.3	(21)	
		MCWB	-0.6	0.7	0.9	4.8	12.4	15.3	17.3	15.9	12.6	7.4	3.7	0.7	(22)	
	5%	DB	-1.6	0.5	2.1	7.9	18.1	20.8	24.4	21.0	15.0	7.4	2.7	0.6	(23)	
		MCWB	-2.0	-0.2	0.1	3.4	11.5	13.3	16.0	15.5	11.7	6.4	2.3	0.2	(24)	
	10%	DB	-3.2	-0.9	1.0	6.1	15.0	18.6	22.3	18.8	13.4	6.2	1.7	-0.2	(25)	
		MCWB	-3.6	-1.6	-0.4	2.3	9.2	12.3	15.6	14.4	11.0	5.4	1.3	-0.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.1	2.2	2.9	8.0	14.6	18.3	20.0	19.5	14.9	10.1	5.7	1.7	(27)	
		MCDB	1.5	3.2	5.7	14.6	21.4	22.4	25.0	25.6	18.0	10.5	5.9	2.0	(28)	
	2%	WB	-0.5	0.9	1.5	5.2	13.1	16.5	18.7	17.5	13.3	7.7	3.9	0.9	(29)	
		MCDB	-0.3	1.6	3.1	9.2	21.4	21.3	23.0	21.3	16.1	8.3	4.3	1.2	(30)	
	5%	WB	-1.9	-0.1	0.6	3.9	11.5	14.8	17.7	16.2	12.2	6.5	2.3	0.2	(31)	
		MCDB	-1.6	0.6	1.7	6.5	17.7	19.5	22.1	19.3	14.3	7.3	2.6	0.6	(32)	
	10%	WB	-3.6	-1.5	-0.3	3.1	9.6	13.2	16.7	15.1	11.4	5.5	1.3	-0.6	(33)	
		MCDB	-3.2	-1.0	0.8	5.7	14.1	17.2	21.1	18.2	13.1	6.2	1.6	-0.3	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	8.6	8.4	12.1	10.7	10.8	9.9	10.2	9.6	8.1	5.0	5.1	6.7	(35)	
		MCDBR	8.0	5.6	9.3	12.6	16.9	14.3	14.1	13.4	11.7	5.7	4.6	4.8	(36)	
	5% WB	MCWBR	7.9	5.2	7.4	7.9	8.7	6.6	6.3	7.1	7.7	4.9	4.5	4.6	(37)	
		MCWBR	8.0	5.7	6.7	10.4	15.6	11.8	11.7	10.8	9.6	5.7	4.5	4.7	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.199	0.265	0.299	0.332	0.346	0.345	0.363	0.341	0.316	0.286	0.352	0.199	(40)	
		taud	1.924	2.092	2.140	2.212	2.427	2.472	2.445	2.478	2.531	2.344	2.680	1.948	(41)	
	Ebn at Noon	Ebn at Noon	210	574	748	823	855	865	834	817	759	570	228	20	(42)	
		Edh at Noon	16	55	87	106	96	94	94	82	62	43	17	3	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	0.08	0.59	2.06	3.84	4.75	5.05	4.69	3.41	1.82	0.59	0.13	0.01	(44)	
	RadStd	RadStd	0.01	0.11	0.20	0.37	0.44	0.36	0.48	0.34	0.19	0.10	0.02	0.00	(45)	

Historical Trends

		DBAvg	Heating		Cooling		Degree-Days			
			99% DB	99% DP	1% DB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (21 neighbors)	+0.58	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page